Regime detection that can say what happened — and a measurement of how much that is worth

An auditable explanation layer for any regime model. BSc Mathematics with Statistics, University of Bristol.

> **The claim.** A two-state HMM refitted inside every walk-forward fold separates SPY volatility regimes in **18 of 20 folds** containing both states (sign test p = 0.00040). Reading only news published *before* each of 20 transitions, a language model produced cited explanations matching back to their own fortnight **72% of the time against 33% chance**, with **96.4% of 474 claims grounded, zero fabricated**. Whether they are *diagnostic* is unresolved: **+18.5pp, 95% CI [-12, 47]pp**.

PROBLEM

A regime model tells you the market changed but not what happened, because it only ever sees returns. Its output is coloured bands a human interprets from memory — undocumented, unauditable, different for everyone who looks.

SOLUTION

The HMM decides *when* the regime changed; the language model only describes *what was in the news* then. No statistic comes from the language model. For each date the tool retrieves a 14-day window that *provably* closes on it, and returns an explanation whose every claim cites a supplied article.

EVIDENCE THE DETECTOR WORKS

Stressed / calm volatility	Same-day	Forward 20d
SPY *(fitted)*	**2.18×**	**1.69×**
Nikkei 225	1.54×	1.18×
Hang Seng	1.46×	1.32×
DAX	1.87×	1.58×

2.18× means stressed days move 2.18 times as much as calm days. Same-day is partly definitional — the HMM is fed trailing volatility; **forward-20d** is volatility that had *not happened* when the state was assigned, and is the honest number. Nikkei at 1.18× is close to nothing and is reported as such.

IMPACT & VALUE

A desk with thirty portfolio managers gets thirty private readings of one regime chart. None is written down, none is auditable, and the reasoning leaves with whoever remembers that week. This replaces "trust me, that band is the Greek referendum" with a written explanation, citing dated sources, carrying a number that says how specific it is.

The reusable asset is the controls, not the explanations. Anyone can ask a model what happened in a given month; nobody can otherwise tell you whether the answer is specific to that fortnight or commentary fitting any month. Point this at any regime model's dates — any asset, any method — and each explanation returns with a blind-match score against chance, a grounding rate against a random-citation floor, and a matched control arm.

USE OF AI

`claude-opus-5` via the Anthropic Messages API — one call per window, effort high, schema enforced server-side. Prompts are version-controlled files with a dated iteration history, never inline strings; every call logs model id, prompt hash and input hash. Built with Claude Code. News via the revision-pinned Wikipedia MediaWiki API, prices via yfinance, statistics via hmmlearn and scikit-learn, tool in Streamlit.

THREE CONTROLS

- **Hindsight.** A window object refuses to construct if any item post-dates it: 60 windows, 10,888 items, zero survivors. Pages are pinned by revision id because a mean of **38.2%** of a current Wikipedia day-page (peak 71.2%) was written *after* the boundary.
- **Memorisation.** Dates stripped; every claim must cite; grounding scored lexically, so the check has no world knowledge.
- **Placebo.** Era-matched controls, identical prompt.

> **The honest bound on that value.** The placebo arm says these explanations are not yet shown to be *diagnostic* of a regime change. So what this buys today is auditability, not signal generation: an undocumented interpretation becomes a documented one, with a figure attached saying how far to trust it. A smaller claim than the obvious one, and unlike the obvious one it is defensible.

REFLECTIONS

What worked: writing the controls before generating anything, including the negative controls that prove the blind-match test can fail on boilerplate. Making the retrieval boundary structural rather than a convention.

What I would change: the placebo test was doomed by design — power at the observed gap is 0.17, and the binding constraint is 20 transitions, not the number of controls, so more placebos could never have helped. Pre-specifying volatility *onsets* at a shorter dwell threshold would roughly triple it. One leak also stays open: the boundary is 23:59 UTC and Wikipedia reports the US close the same evening, though controls carry *more* of it than transitions, so it does not manufacture the result.